nature reviews electrical engineering

https://doi.org/10.1038/s44287-024-00126-6

Review article

Check for updates

Electrochemical impedance spectroscopy in vivo for neurotechnology and bioelectronics

Brittany Hazelgrove[1], Lukas Matter[2], Brad Raos[1], Bruce Harland[1], Leo Cheng[3], Maria Asplund[2] & Darren Svirskis[1]

Abstract

Electrochemical impedance spectroscopy (EIS) is a well-established electrochemical technique that provides invaluable information regarding the properties and functionality of electrodes within bioelectronic devices. EIS is the primary technique that reports on electrode properties in vivo using the implanted device itself. Nevertheless, there are many inconsistencies in the way this technique is implemented and reported on. Without a clear understanding of the experiment and experimental set-up, it is challenging to draw meaningful conclusions and for results to be extrapolated across studies to benefit and advance the field. This Review discusses in vivo EIS experiments, specifically focusing on challenges in the experimental set-up, the equipment used, data presentation and circuit modelling for neural interfaces. We propose guidelines for methodical reporting and a consistent, standardized use of terminology, paramount in understanding the performance of electrodes functioning at neural interfaces and promoting the transferability of findings across studies.

Sections

Introduction

Electrochemical impedance spectroscopy

Recording impedance spectra in vivo

Interpretation of impedance spectra recorded in vivo

Outlook

[1]School of Pharmacy, Faculty of Medical and Health Sciences, University of Auckland, Auckland, New Zealand. [2]Department of Microtechnology and Nanoscience, Chalmers University of Technology, Gothenburg, Sweden. [3]Auckland Bioengineering Institute, University of Auckland, Auckland, New Zealand. e-mail: brittany.hazelgrove@auckland.ac.nz; maria.asplund@chalmers.se; d.svirskis@auckland.ac.nz

Key points

- Electrochemical impedance spectroscopy is a valuable tool for evaluating the stability of electrodes at a neural interface.
- Consideration of the equipment limitations and electrochemical set-up (electrode configuration and locations) is critical for meaningful interpretation of impedance measurements.
- Caution is advised when drawing conclusions from a single frequency measurement. It is recommended that the linearity assumption is supported by an accompanying full-spectrum measurement or a Lissajous plot.
- Models used in the electrical circuit modelling of impedance spectra should be validated and all elements should represent physical properties at the electrode interface.
- The development of an on-the-probe reference electrode would enable more robust in vivo measurements and the ability to separate electrode surface changes from tissue impedance.

Introduction

Electrochemical impedance spectroscopy (EIS) is the primary technique that enables the assessment of electrode properties and functionality in vivo, using the implanted device itself, without any additional interventions. Thus, it is invaluable in the field of bioelectronics and neural implants. Implantable bioelectronics encompasses electronic devices that interact with biological interfaces, such as cochlear implants, pacemakers and visual prosthetics. Neural interfaces represent a subset of bioelectronics that interact specifically with the nervous system, principally through recording from or stimulating neurons[1]. In general, EIS provides a metric to assess electrode functionality and tissue behaviour and supports an understanding of the failure modes of recording and stimulating electrodes. Historically, impedance spectroscopy has been used to study a wide range of interfacial phenomena such as corrosion kinetics, diverse surface reactions, electrochemical deposition and the synthesis of diverse materials[2,3]. The two main categories of impedance spectroscopy have been described as EIS and everything else[4]. The difference is that EIS specifically refers to the analysis of solid and liquid systems where conduction between electrodes is predominantly ionic[5,6]. This technique enables real-time, non-destructive and label-free measurements of electrode performance and various cell processes such as biofouling, cell adhesion, proliferation and apoptosis[7]. The 'everything else' category includes analysis of materials such as dielectrics, where conduction mechanisms are predominantly non-ionic.

EIS is recorded by setting up an electrochemical cell and recording the response of an electrode of interest to a voltage or current perturbation, usually a sine wave. Features of the response, such as damping or a phase shift of the sine wave, represent different physical processes occurring at the interface of the electrode and the electrolyte/surroundings. Further information can be gathered by carrying out this analysis across the frequency spectrum and modelling the system using an equivalent electrical circuit, with each reaction at the interface represented as an electrical circuit element. It is common for researchers to report in vivo EIS data to complement electrophysiology, and empirically, low impedance in vivo has been found to correlate with high-quality neural recordings. However, a clear consensus on reporting and interpreting EIS data is missing. Furthermore, the empirically founded interpretations may not always align with the electrochemist's point of view on EIS.

In this Review, we focus on EIS as an in vivo technique and provide guidance for implantable bioelectronic systems at the neural interface. We introduce the theory of the experiment, considerations for recording impedance spectra in vivo and the interpretation of the recorded spectra. We highlight the challenges introduced when moving away from the laboratory bench and towards electrodes and the in vivo environment. Finally, we present the role of EIS in understanding the electrode–tissue interface and the relevance and impact of the foreign body response (FBR) to neural implants. With the explosion in research efforts towards functional neural implants, this Review is aimed at researchers and clinicians working with implantable neural interfaces, such as brain–machine interfaces, neural recording devices, deep brain stimulation electrodes and spinal cord stimulators, supporting readers to extract meaningful data and draw robust conclusions using EIS within in vivo settings.

Electrochemical impedance spectroscopy

Impedance spectroscopy is a classical electrochemical method, typically simple to conduct but complex and challenging to interpret. Here, we aim to provide background to the impedance experiment, starting with electrochemistry basics.

The electrochemical experiment

The impedance measurement technique consists of applying an electrical stimulus, typically a sine wave at a given frequency, across the electrochemical cell and recording the output response. The output magnitude and phase with respect to the input signal represents the impedance of the electrode and electrolyte and any reactions occurring at the electrode–electrolyte interface[8]. Interfacial phenomena that can be studied using impedance include corrosion kinetics, surface reactions, electrochemical deposition and the synthesis of diverse materials, such as in electrode development[5].

The impedance experiment is carried out using either a potentiostatic method, which refers to the application of a defined voltage sine wave/perturbation with the measurement of the current response, or a galvanostatic method, with the application of a current sine wave/perturbation and measurement of the voltage response[9].

There are three key criteria that must be met for accuracy and confidence in an impedance measurement: linearity, stability and causality[9]. Linearity implies that the recorded output signal contains the same frequency as the input signal. Stability, sometimes referred to as time invariance, means that the system should be in a steady state. Two measurements taken immediately after one another should give the same outcome. The system should also obey causality, where the recorded response exists solely due to the applied input signal[9,10]. The Kramers–Kronig relations are a set of equations that have been developed to validate the linearity, stability and causality of impedance measurements[11]. The Kramers–Kronig relations are built from the concept that there is interdependence between the real (Z') and imaginary (Z'') components, with the real part calculated from the imaginary in Eq. 1 and the imaginary as a function of the real part in Eq. 2:

$$Z'(\omega) = R_\infty + \frac{2}{\pi}\int_0^\infty \frac{xZ''(x) - \omega Z''(\omega)}{x^2 - \omega^2}\mathrm{d}x \quad (1)$$

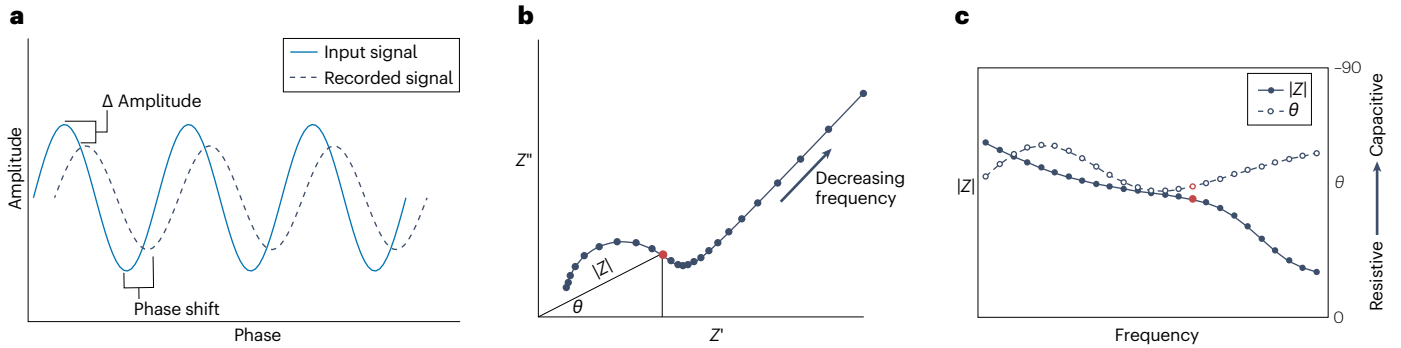

Fig. 1 | The relationship between the Nyquist and Bode representations of impedance spectra. a, Visual representation of the amplitude and phase response, the key parameters that define the impedance response. **b**, Nyquist representation, where Z' represents the real component of impedance and Z'' the imaginary component, schematically showing the semicircular arc seen in in vivo impedance spectra[17,22]. Each data point represents the impedance recorded at a given frequency, from 1 Hz to 100 kHz. The Pythagorean relationship between Z', Z'', magnitude of impedance ($|Z|$) and phase angle (θ) centred around an example data point (orange). **c**, Bode representation. Orange data points are calculated as shown in the inset in **b**.

$$Z''(\omega) = \frac{2\omega}{\pi} \int_0^{\infty} \frac{Z'(x) - Z'(\omega)}{x^2 - \omega^2} \mathrm{d}x \quad (2)$$

where ω is the angular frequency, R_∞ is the high frequency limit of the impedance, x represents the limits of integration, Z' is the real component and Z'' is the imaginary component. The frequencies that meet these conditions, where the real and imaginary parts can be calculated in this way, fulfil the requirements for a meaningful impedance measurement.

Provided the equipment limits are not reached, and the assumptions above are met, both potentiostatic and galvanostatic methods should give as output the same impedance response. The impedance response is frequency-dependent, which is why the electrode–electrolyte interface impedance is recorded across a frequency spectrum and is the rationale behind the term 'spectroscopy'. The impedance can be visually understood at each frequency as a combination of an amplitude reduction and a phase shift. Changes in the amplitude of the recorded signal represent the impedance magnitude ($|Z|$), whereas the time delay represents the phase angle (θ) of the impedance (Fig. 1a).

Graphical representation of impedance spectra

Impedance is represented by complex numbers — comprising a real and an imaginary component. The real component (Z') represents the resistive contributions, whereas the imaginary component (Z'') comprises the reactance of the system. As with any complex number, impedance can also be presented as a polar vector ($|Z|$, θ), calculated from the real and imaginary components (see equations 1 and 2):

$$|Z| = \sqrt{(Z')^2 + (Z'')^2} \quad (3)$$

$$\theta = \tan^{-1}\left(\frac{Z''}{Z'}\right) \quad (4)$$

The Nyquist plot (Z' versus Z'') (Fig. 1b) and the Bode plot ($|Z|$ and θ versus frequency (f)) (Fig. 1c) are the two main methods to present impedance spectra. Both representations are interchangeable, but each highlights different aspects of electrode frequency dynamics. Although shifts of the real and imaginary parts are easily detectable in the Nyquist representation, the frequency response is not explicitly indicated[4,10,12]. The larger impedance recorded at low frequencies determines the scale of the axes, resulting in a disproportionate presentation of the complex impedance at low frequencies, with high-frequency responses potentially hidden[10] (Fig. 1b).

In contrast, in Bode plots the log scale enables trends across the entire spectrum to be quickly identified. This representation highlights the effect of frequency on the reactance and resistance of the interface. However, the relative contribution of resistance versus reactance is less intuitive to discern compared with the Nyquist plot[10]. The Bode representation is especially helpful to identify the high-pass filter characteristics of the electrode; that is, to estimate whether the signal frequency to be recorded is in the optimal high-pass region of the specific electrode, where only resistive attenuation applies. When presenting data in the Bode format it is important to make clear the θ value. A capacitive phase has an angle of −90°, whereas a purely resistive phase has an angle of 0°. The phase axis can either be presented from −90° to 0° or as the negative of the phase from 0° to 90° (Fig. 1c).

Nyquist and Bode representations contain the same information, just visualized in different ways. Datasets can be further analysed using electrical circuit modelling to extract key information relating to specific frequencies or processes at the interface.

Electrical circuit modelling of the electrode interface

Electrical circuit modelling of EIS data is valuable to interrogate the contributions from the electrode–electrolyte interface and the surrounding electrolyte, and to analyse the mechanisms contributing to the reactance and resistance of the system. The charge-transfer and mass-transfer mechanisms at the electrode and within the bulk electrolyte solution can be represented by an electrical circuit element and combined to form an equivalent circuit model of the system[13]. Circuit modelling is an extensive research topic[10]. Here, we focus on its practical application for electrode diagnostics and performance.

The most common circuit model of an electrochemical cell is the simplified Randles circuit (Fig. 2a), where the electrode–electrolyte interface is represented as a parallel circuit containing a resistor (charge transfer resistance R_{ct}) and Warburg impedance (Z_W), in parallel with a capacitor, the double-layer capacitance (C_{dl}): $[(R_{ct}Z_W)||C_{dl}]$. A second resistor, representing the resistance of the surrounding electrolyte

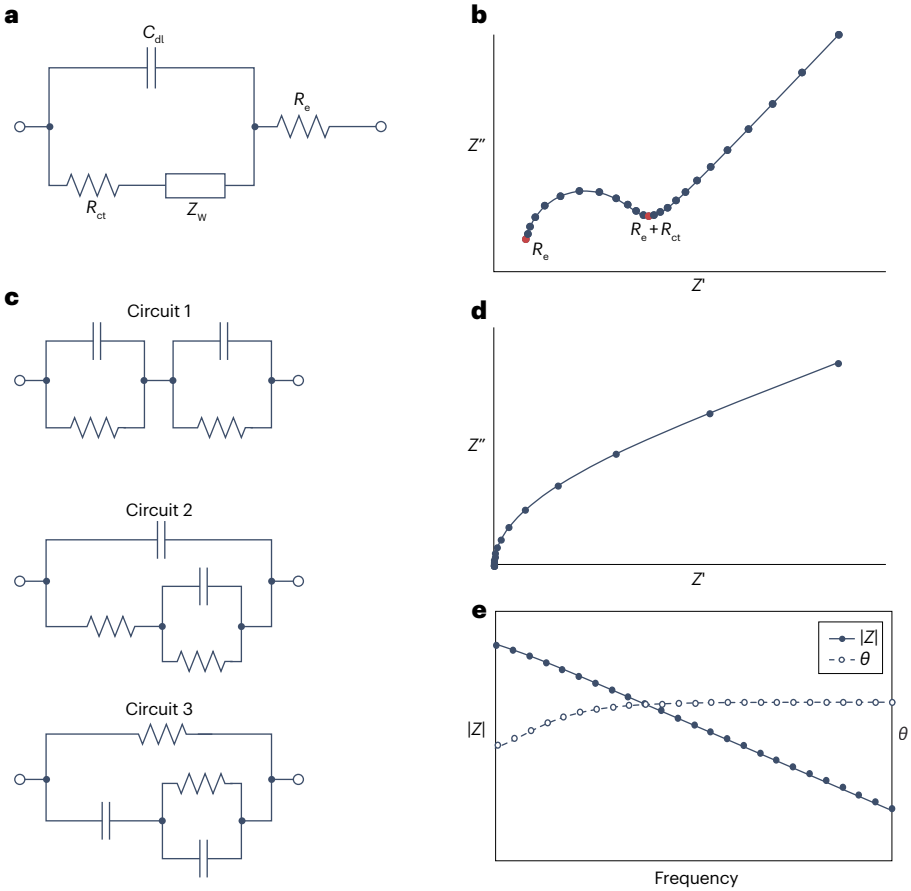

Fig. 2 | Circuit modelling of impedance spectroscopy. a,**b**, Randles circuit representation of impedance spectra through the circuit model (**a**) and the Nyquist representation (**b**). In **a**, the parallel component represents the electrode–electrolyte interface. **c**–**e**, Examples of circuit models, each comprising two capacitors and two resistors in a unique arrangement (**c**), that can be fitted to the same impedance spectra as a Nyquist plot (**d**) and a Bode plot (**e**)[4]. C_{dl}, double-layer capacitance; R_{ct}, charge transfer resistance; R_e, resistance of the electrolyte or the tissue in the in vivo case[10]; $|Z|$, magnitude of impedance; Z', real component of impedance; Z'', imaginary component of impedance; Z_W, Warburg impedance to model the diffusive properties; θ, phase angle.

(R_e), is in series with $(R_{ct}Z_W)||C_{dl}$. The parallel component, $(R_{ct}Z_W)||C_{dl}$, represents two pathways; the charging and discharging of the double-layer capacitance, represented by C_{dl}; and the faradaic process, represented by R_{ct} and Z_W. Z_W results from the frequency dependence of a diffusion mechanism at the electrode–electrolyte interface[13]. At high frequencies, Z_W is negligible, as on the short timescale the influence of diffusion on the current is small.

By fitting the impedance of the electrode with the Randles circuit, the various mechanisms that contribute to the electrode charge transfer impedance are visualized. From a least square fit, the magnitude of C_{dl}, R_{ct} and R_e can be extracted from the complex data. At high frequencies, the impedance across a capacitor approaches zero, meaning that the C_{dl} pathway in the Randles circuit can be approximated as a short circuit and the impedance can be represented solely by R_e (Fig. 2b). When analysing identical electrodes in different concentrations of conductive electrolyte, it can be expected that only the impedance response at high frequencies changes[14]. If the electrode is porous to the electrolyte, the result will deviate from this idealistic model[15,16].

Even though physical processes at the electrochemical interface can be represented by solid-state capacitors and resistors, this is a simplification which is appropriate only within certain boundary conditions for which the correctness of the models has been experimentally validated. The Randles circuit is a well-validated model that represents an electrode interface and can include variations of the Warburg element to represent electrode coatings that add complexity in the interfacial processes. However, the model covers only the interactions at the electrode surface, assuming the electrolyte is a purely resistive pathway. When considering impedance analysis of neural interfacing electrodes in vivo, there are further complexities added to the system, such as the non-homogeneous nature of biological tissue, where a single electrolyte resistance (R_e) might be insufficient.

When fitting a circuit model to impedance data, the goodness of fit of the model needs to be evaluated before interpreting the parameter values of the model (for example, that changes in R_e over time represent changes in the tissue response). By quantifying the goodness of fit, it is possible to investigate and quantify the suitability of different models, which comprise different elements[17], to fit a specific dataset. At a basic level, users should qualitatively examine residual plots for latent trends or plot both the data and the model fit on the same axes to provide visual confidence in the chosen model[18,19]. Additionally, when fitting a circuit model to impedance spectra one must consider the error structure of the measurement. The stochastic error can be reduced through averaging multiple cycles at each frequency, particularly effective at high frequencies[9]. Evaluation of the assumptions of linearity, stability and causality for each frequency is also encouraged through assessing the Kramers–Kronig relations for each frequency.

Regardless of the application, care must be taken when using circuit modelling to explain reactions and phenomena. For a given impedance spectrum, there is no unique circuit model[10]. An equivalent circuit model using the same elements may be connected in different ways but still exhibit the same response[4]. For example, three electrical circuits, each consisting of two resistances and two capacitances, fit the

same impedance spectra (Fig. 2c–e). This is a limitation when modelling impedance spectra. Thus, it is fundamental to carefully evaluate which underlying physical processes are being modelled and whether the electrical components selected are appropriate representations of real phenomena in the given application. In general, it is recommended to first model the system with the simplest possible circuit, such as the Randles circuit, and then to add further elements and complexity if the response is insufficiently modelled. The additional circuit elements should deliberately represent physical electrochemical processes, and, in the ideal case, the use of the model should be validated in an equivalent physical system where the model components can be varied in a controlled manner. Electrical circuit modelling of the in vivo environment has additional pitfalls (Supplementary Information, section 1).

Two-electrode and three-electrode set-ups of the electrochemical cell

The most common impedance measurement set-up consists of three electrodes that make up the electrochemical cell; the working electrode (WE), the counter electrode (CE) and the reference electrode (RE) (Fig. 3a). The three electrodes are immersed in a common electrolyte, which provides the electrical connection. The electrodes are at an interface where electronic charge carriers in the metal wires are transferred into ionic charge carriers in the electrolyte.

Typically, EIS aims to measure the impedance of the WE interface to the liquid; however, this requires a closed circuit. The CE completes the circuit. In practice, the measured impedance is not solely from the WE but is the sum of impedances from the WE, CE and electrolyte. Assuming both WE and CE are of similar size (Fig. 3b), each will consist of an electrode–electrolyte interface having an equal contribution to the recorded impedance. In this set-up it is challenging to isolate the behaviour of the WE. However, by using a much larger CE, the contribution from the CE interface can be assumed to be negligible compared with the contribution from the WE (Fig. 3c). A common rule of thumb would be to suggest an area ratio of at least 1:10. In practice, the appropriate ratio will depend on multiple factors such as the electrochemically active area of the WE (often enlarged by dedicated low impedance coatings) and the CE interface impedance, expected to increase over time by processes such as biofouling or corrosion. Based on this, the user should make an estimation of the expected proportional contribution of the CE over time and decide whether this would be a relevant influence on the overall EIS spectrum.

To further reduce the influence of the CE and the electrolyte on the measured impedance, a third electrode, the RE, is introduced into the set-up. Addition of the RE allows the contributions from the WE and CE to be separated. The RE provides a known reference potential, making it possible to deduce the impedance representing the WE by the voltage drop across the WE, recorded as the difference in the applied voltage and reference potential (Fig. 3a). Care must be taken to ensure that the RE is non-polarizable. Polarization of an electrode occurs when the charge that forms on the surface is not immediately converted into current but builds up as a polarization potential at the electrode. For a RE, this can be detrimental, with the effect being a time-varying reference potential, fluctuating with the charge build-up, decreasing the reliability of the impedance measurement. A commonly used RE for benchtop measurements is the Ag/AgCl electrode, which is a non-polarizable electrode system with a defined potential. Ideally, any voltage imposed on the AgCl coating will immediately result in a proportional dissolution or deposition of AgCl salt. The reference potential is further stabilized by using a liquid junction, where the electrolyte in contact with the AgCl wire is consistently defined, regardless of which electrolyte is chosen for the measurement cell[20].

For in vitro set-ups, electrolytes such as phosphate-buffered saline (PBS) that mimic the osmolarity and ionic composition of the human body (such as the cerebrospinal fluid) are typically employed. For in vivo applications, the tissue is the electrolyte, with a much more complex, non-homogeneous and changing composition that includes various proteins.

Recording impedance spectra in vivo

Although EIS is an established technique, there are many challenges in its translation into the in vivo environment. There are challenges faced in both the set-up of the electrochemical cell (CE and RE materials and locations) and in understanding the limitations of the electrophysiology equipment used.

Considerations and limitations of the in vivo electrochemical cell

In vivo EIS can be applied to evaluate the status of recording and stimulating electrodes and to assess the impedance of the tissue[21,22]. Translating the in vitro electrochemical set-up into an in vivo experiment is, in principle, simple, but in practice entails challenges to ensure the validity of the measurements and their interpretation. For in vivo applications, the electrochemical cell must be miniaturized due to space constraints and to minimize the invasiveness of implantation and removal procedures.

Various devices can be implanted to interface with the brain[23] (Table 1), including needle arrays[24–29], flexible and rigid shanks[17,19,30–34], microwire arrays[27,35–42] or surface implants[43]. Electrode materials used in these studies include iridium oxide (IrOx) and iridium[24,25,30,31,33–35], tungsten[22,38,42], platinum[19,25,26,28,29,35,36,43], platinum–iridium (PtIr)[17,32,37,39,40], stainless steel[41] and poly(3,4-ethylenedioxythiophene) (PEDOT)-based polymers[30,32,33]. Additionally, there is variability in implantation positions, with the most frequent being the motor cortex[24,27–30,35,40,41,43,44], somatosensory cortex[22,35,38,39,42] and visual cortex[26,27]. Animals involved in the vivo models include rats[19,22,24–27,30–34,37–44], mice[36], cats[45], monkeys[17,28,29] or humans[35].

The most measured and reported representation of EIS is a single measurement at 1 kHz[31]. Although single-frequency impedance measurements might enable fast detection of electrode failure, a complete frequency spectrum is needed to assess the tissue's changed electrical properties. Frequency spectra are often collected in the hertz to kilohertz range, with megahertz data occasionally reported.

Most in vivo studies use a two-electrode set-up for EIS, with the CE most often consisting of a single stainless steel bone screw in the skull[19,22,27,30–32,34,36–39,41,42,44] or a stripped wire[17,25,29,40]. Sometimes, more than one skull screw is connected to the CE[32,37], adding redundancy in case the connection to one bone screw fails. A skull screw or stripped wire usually has a significantly larger area than the WE microelectrode, thus minimizing its share of the impedance (Fig. 3d). A contrasting approach is to use a CE integrated on the micro-sized implant[33,46]. When the CE is integrated within the implant, the tissue volume between the CE and WE is reduced, decreasing the tissue contribution to the measured impedance. However, the integrated CE is inevitably going to be smaller than a skull screw and therefore adds more impedance to the EIS (Fig. 3e).

In vivo EIS can also be measured with a three-electrode set-up, using an implantable RE[24,26,33,36]. It is important to note that the term RE is also commonly used in electrophysiology measurements to note

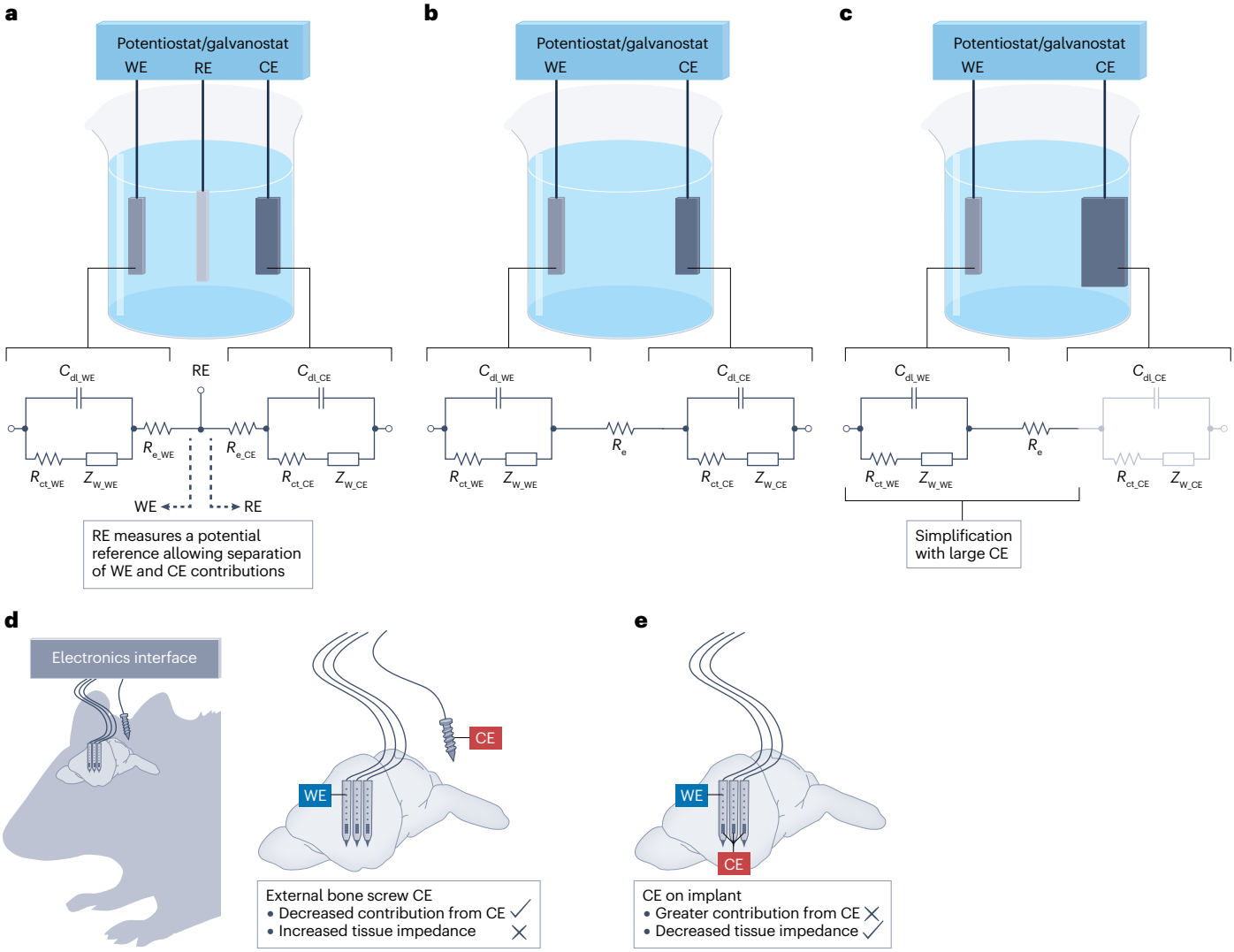

Fig. 3 | Overview of two-electrode and three-electrode set-ups for recording impedance spectra in vitro and in vivo. a–**e**, Two-electrode and three-electrode set-ups for recording impedance spectra in vitro (**a**–**c**) and in vivo (**d**,**e**). **a**, Three-electrode electrochemical cell comprising a working electrode (WE), reference electrode (RE) and counter electrode (CE). Beneath the electrochemical cell is the Randles circuit representation of the impedance experiment. **b**, Two-electrode electrochemical cell, consisting of a WE and CE. The circuit model highlights that contributions from the WE and CE both contribute to the recorded impedance. Without the RE, R_e represents the electrolyte resistance between the WE and CE. **c**, Two-electrode electrochemical cell, as seen in **b**, but with a much larger CE. The larger CE simplifies the response, assuming negligible impedance contribution from the CE. **d**, An electrochemical cell with a bone screw as a CE, located further from the implant, decreasing the CE contribution while increasing the tissue volume between the WE and CE and increasing the tissue's contribution to the recorded impedance. **e**, An electrochemical cell where the CE is on the implant, reducing the influence of the tissue on the recorded impedance but increasing the CE contribution. C_{dl_WE} and C_{dl_CE}, double-layer capacitance at the WE and CE, respectively; R_{ct_WE} and R_{ct_CE}, charge transfer resistance at the WE and CE, respectively; Z_{W_WE} and Z_{W_CE}, Warburg element representing diffusion mechanisms at the electrode–electrolyte interface; R_{e_WE} and R_{e_CE}, electrolyte resistance between the WE and RE and between the CE and RE.

the reference signal to the differential amplifier. In general, the CE is not suitable as a RE because it is active in the charge transfer, which causes polarization, meaning the reference potential might drift over time. Thus, an additional electrode is necessary. The RE can be either positioned far from the WE (skull screw[26,36], tail[45] and leg[24]) or integrated within the implant[33]. In electrochemistry, it is common to position the RE close to the WE to be able to measure the potential of the WE as accurately as possible. This is facilitated with a RE on the implant itself.

The electrochemical set-up determines the impact of the tissue on the measured impedance. In the three-electrode set-up, the tissue volume between the WE and RE contributes to the impedance. In the two-electrode set-up, the tissue volume between the WE and CE impacts the impedance. With two electrodes, the applied amplitude of the sinusoidal signal is constant. In contrast, in the three-electrode configuration the applied signal amplitude can be increased up to the point where the specified set amplitude (for example, sinusoidal

Table 1 | Key parameters of in vivo electrochemical impedance spectroscopy studies

Animal	Implant location	Implant type	EIS set-up	Frequency	Amplitude	Ref.
Two-electrode set-up						
Rat	Cortex	Polyimide shank with IrOx electrodes	CE: SSSS	10 Hz–100 kHz	10 mV RMS	34
Human	Motor and somatosensory cortex	Needle array with platinum and IrOx electrodes	CE: not specified	1 kHz	10 nA peak to peak[a]	35
Rat	Somatosensory cortex	ECoG with platinum, IrOx and PEDOT:PSS electrodes	CE: SSSS	1 Hz–100 kHz	100 mV RMS	92
Rat	Motor cortex	ECoG with platinum electrodes	CE: titanium cranial screw	1 Hz–1 MHz	50 mV	43
Rat	Substantia nigra pars reticulata	PtIr microwire array	CE: 2× SSSS	1 kHz[a]	Not specified	37
Rat	Subthalamic nucleus	PtIr microwire; tips coated with PEDOT:BF4	CE: 4× SSSS	1 Hz–100 kHz	10 mV	32
Rat	Sciatic nerve	Needle array with IrOx and platinum electrodes	CE: wire inside muscle	1 Hz–1 MHz	Not specified	25
Monkey	Motor cortex	Needle array with platinum electrodes	CE: not specified	1 kHz[a] 100–2,050 Hz	Not specified	28
Rat	Somatosensory cortex	Tungsten microwire array	CE: SSSS	1 Hz–2 kHz	4 mV peak to peak	38
Rat	Somatosensory cortex	PtIr microwire array	CE: SSSS	1 Hz–2 kHz	4 mV peak to peak	39
Rat	Motor cortex	PtIr microwire array	CE: platinum wire in space between bone and muscle	1 kHz[a] 1 Hz–100 kHz	Not specified	40
Monkey	Motor and premotor cortex	Needle array with platinum electrodes	CE: subdural wire	1 kHz[a]	50 pA peak to peak	29
Rat	Somatosensory cortex	Tungsten microwire array	CE: SSSS	1 Hz–2 kHz	4 mV peak to peak	42
Rat	Primary motor cortex	Polyimide shanks with platinum electrodes	CE: SSSS	100 Hz–1 MHz	100 mV	19
Monkey	Thalamus and subthalamic nucleus	Polyurethane shank with PtIr contacts	CE: coiled Ag/AgCl wire in contralateral access chamber submerged in 0.9% NaCl	0.5 Hz–10 kHz	25 mV RMS	17
Rat	Motor cortex	Silicon shank with iridium electrodes	CE: SSSS	10 Hz–10 kHz	25 mV RMS	44
Rat	Motor cortex	Stainless steel microwire array	CE: SSSS	100 Hz–1 MHz	10 mV	41
Rat	Visual and motor cortex	Various: needle array, silicon shank, microwire array	CE: SSSS	0.1 Hz–10 kHz	Not specified	27
Rat	Somatosensory cortex	Tungsten microwire array	CE: SSSS	100 Hz–2 kHz	5 mV	22
Rat	Motor cortex	Silicon shank with iridium and PEDOT electrodes	CE: SSSS	10 Hz–10 kHz	25 mV RMS	30
Rat	Cerebral cortex	Silicon shank with iridium electrodes	CE: SSSS	1 kHz	5 mV	31
Three-electrode set-up						
Mouse	Hippocampus	Platinum nanoparticle-coated microwire array	CE: SSSS RE: SSSS	1 Hz–100 kHz	Not specified	36
Rat	Motor cortex	Needle array with IrOx tips	CE: platinum wire at tail RE: Ag/AgCl at tail	1 Hz–100 kHz	10 mV RMS	24
Rat	Visual cortex	Needle array with platinum electrodes	CE: SSSS ipsilateral RE: SSSS contralateral	10 Hz–32 kHz	10 mV RMS	26
Rat	Hippocampus	Polyimide shank with IrOx PEDOT:DEX and PEDOT:PSS electrodes	CE: implant-integrated platinum RE: implant-integrated IrOx	1 Hz–10 kHz	10 mV	33
Cat	Cortex	Needle array with IrOx tips	CE: AgCl foil at the leg RE: platinum mesh at the other leg	1 Hz–100 kHz	10 mV RMS	45

Implant locations, implant types, cell set-up and key experimental parameters are reported. CE, counter electrode; ECoG, electrocorticography; EIS, electrochemical impedance spectroscopy; IrOx, iridium oxide; PEDOT, poly(3,4-ethylenedioxythiophene); PEDOT:BF_4, tetrafluoroborate-doped poly(3,4-ethylenedioxythiophene); PEDOT:DEX, dexamethasone-doped poly(3,4-ethylenedioxythiophene); PEDOT:PSS, poly(3,4-ethylenedioxythiophene) polystyrene sulfonate; PtIr, platinum–iridium; RE, reference electrode; RMS, root mean square; SSSS, stainless steel skull screw. [a]Galvanostatic control.

10 mV root mean square (RMS)) between the WE and RE is reached. The applied signal in vivo might need to be adapted with time to account for an increased impedance due to tissue encapsulation or electrode polarization, which the three-electrode set-up accounts for, resulting in a higher recorded current than the two-electrode set-up[9]. Although there are many advantages to a three-electrode set-up, there are many challenges with implementing an effective RE in vivo.

The use of a RE is especially important in applications which require a precise measurement of the potential of the WE. This is the case, for example, of in vivo drug delivery or during the detection and identification of analytes with fast-scan cyclic voltammetry[20,47]. Standard electrochemical REs cannot be readily used in vivo because they might incorporate toxic chemicals (such as mercury in the calomel electrode) or require an inert environment (for example, constant H_2 pressure in the hydrogen electrode). The basis for a RE to produce a stable and predictable potential is that the electrolyte environment is controlled. Even if this is not fully possible in tissue, the environment is still largely predictable, as the salt concentration and pH are closely controlled by the body's own regulatory system. For this reason, a chlorinated silver wire, such as a silver wire coated with AgCl, can be used to secure stable reference potentials under certain conditions, including the presence of a fixed concentration of chloride ions (Cl^-) in the surrounding environment, the absence of tissue encapsulation and the fact that the electrode does not get polarized by an external field. However, this electrode lacks the required long-term stability in vivo and leads to silver ions, which are known to be cytotoxic[48–50], being eluted into the tissue over time. A viable option is to replace the silver wire with another material which does not dissolve (such as IrOx, boron-doped polycrystalline diamond), or to temporarily insert an Ag/AgCl RE embedded in a buffering layer of gel, to prolong the time until dissolution has notable consequences[51]. For example, an Ag/AgCl RE mounted on the rat's tail was used for EIS[24]. The distance between the WE in the brain and the RE at the tail demands careful evaluation of the measured potential of the WE. A compensation of R_e (iR drop compensation) might be necessary for better accuracy[52].

Another option for in vivo REs is to use biocompatible and electroactive materials that are stable over time[47,53,54]. Although the reference potential might drift over long timescales, the value is likely to remain stable over each measurement. A practical example is the use of IrOx, which can be deposited as a thin-film electrode, a concept first proposed in the biosensor community[53] and later adopted as a neural on-probe RE[33]. IrOx is expected to have a stable open cell potential under a stable pH, a condition likely to be fulfilled within the time span of a measurement. Different hydration states of the film could lead to a drift in open cell potential over time, but as an implanted electrode it can be expected that a steady state will be reached as the electrode remains in water. Although IrOx electrodes are capacitive in nature, their capacitance mechanisms are a mix of double-layer capacitance and pseudo-capacitance. The latter behaviour enables the electrode to absorb or expel any excess charge from/to solution, so that polarization that would change the open cell potential is unlikely to build up[33]. An IrOx RE is more specifically a pseudo-RE, defined as "a metal wire, Ag or Pt, used with the expectation that in experiments where there is essentially no change in the bulk solution, the potential of this wire, although unknown, will not change during a series of measurements"[55].

Equipment and methods used to collect impedance measurements in vivo

Alongside the set-up of the electrochemical cell, there are also considerations on the equipment used to collect EIS measurements. The accuracy of an impedance measurement depends on the environment (for example, ambient noise), appropriate parameters (if the configuration is potentiostatic or galvanostatic) and the equipment used. All impedance spectroscopy equipment has limitations, and output values cannot be blindly trusted.

The important assumption of linearity must be met; the recorded signal must contain the same frequency as the input signal. If the input signal is a sinus of a certain amplitude and phase, the output can vary in amplitude and phase but still needs to be a sinus of the same frequency. One way to test the output signal is to incorporate an oscilloscope into the system to provide real-time monitoring of the time domain signals and validate that the recorded waveform truly is a linear translation of the input waveform. A useful plot that can be produced is the Lissajous plot, where the input waveform is plotted against the recorded waveform. Here, the signal is deemed to be linear when the plot is elliptic and symmetrical[56,57]. If a linear impedance measurement is presented, the Lissajous plot should show a symmetrical output (Fig. 4a). In contrast, for a non-linear measurement, the recorded signal is not a linear translation of the applied signal (Fig. 4b). This type of visualization makes the interpretation of successful versus failed measurements straightforward, making the latter easily detectable. Another method to assess linearity, alongside the assumptions of stability and causality, are the Kramers–Kronig relations[11]. As with the Lissajous analysis, the Kramers–Kronig equations are built into many EIS analysis systems.

When moving from in vitro EIS to in vivo EIS, there is an increase in the impedance of the system and an accompanying increase in the noise sources[58], such as from movement artefacts and/or electrical noise from biological systems. The increased impedance leads to a lower measurement signal, particularly at lower frequencies, causing greater errors and increased contribution from noise sources. When recording EIS, additional components might be required, such as additional connectors and cables, which can also increase the noise contribution to the measurement. To reduce the effects of noise, particularly at low frequencies, the length of the cables should be made as short as possible. Cables should be shielded, and, if possible, the use of a Faraday cage is recommended. The amplitude of the applied measurement signal can also be increased to compensate for larger impedances. As EIS is a linear technique, increasing the input amplitude will increase the recorded signal. However, care must be taken to ensure that the signal is not increased to the point of causing irreversible reactions at the electrodes, defined by the water window (−0.6 V to 0.9 V for Ag/AgCl electrodes), with exact boundaries dependent on the electrode materials, or substantially changing aspects of the measured materials such as the oxidation state of a conducting polymer[16,59]. Additionally, as the input amplitude increases, the linearity assumption is no longer met[60] (Fig. 4c). Different electrochemical effects come into play at the electrode–electrolyte interface, depending on the voltage imposed on the interface. The higher voltage makes additional electrochemical reactions available, which influence charge transfer and, thereby, the output signal in a non-linear way.

The precision of the equipment itself has a major influence on the quality of the measurement. Whereas dedicated electrochemistry equipment often has broad operational ranges and inbuilt functionality to display time-domain data (such as Lissajous plots), measurements in vivo are commonly made with more simplistic instrumentation. Impedance testing equipment that comes packaged within electrophysiology recording equipment is commonly used (Blackrock Neurotech, Intan Technologies). Although these systems are convenient for electrode-function tests, it is important to recognize their limitations in function and precision.

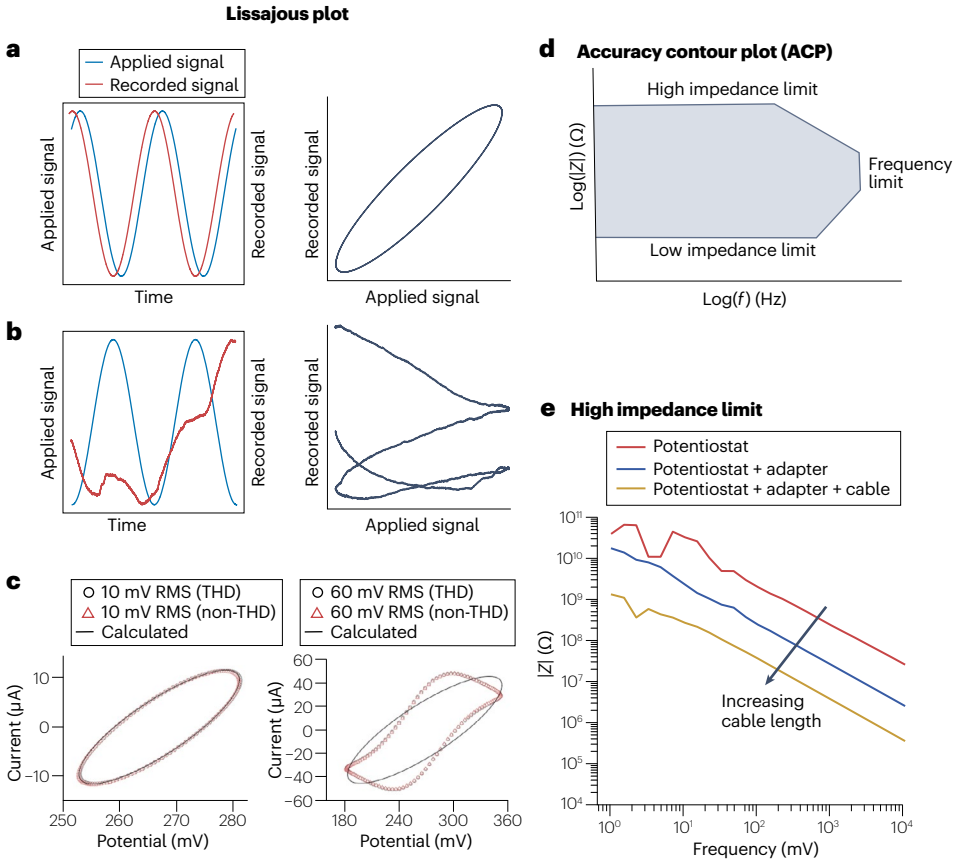

Fig. 4 | Considerations in the equipment used to collect electrochemical impedance spectroscopy data, presented as Lissajous plots and accuracy contour plots. a–**e**, Electrochemical impedance spectroscopy (EIS) data presented as Lissajous plots (**a**–**c**) and accuracy contour plots (ACPs) (**d**,**e**). **a**, Raw data, with the applied signal (blue) and the recorded signal (red) and the corresponding Lissajous plot, where the applied and recorded signals are plotted against each other. The linearity condition is met for this frequency. **b**, Raw data, with the applied signal (blue) and the recorded signal (red) and the corresponding Lissajous plot, where the applied and recorded signals are plotted against each other. The linearity condition is not met in this case. **c**, The effect of increasing input amplitude on the linearity condition, with Lissajous plots for a 10 mV and 60 mV input voltage data. Part **c** is adapted from ref. 60, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/). **d**, Theoretical ACP, highlighting the concept of a bounded region on Bode axes where the recorded impedance will be accurate. **e**, Dataset highlighting the effect of cable length on the upper limit of the ACP. The red line indicates the open circuit potential of the potentiostat, blue line the open circuit potential of the potentiostat and an adapter required for the measurements, and yellow line the open circuit potential of the potentiostat, the adapter and an additional cable necessary for recording EIS data in vivo. THD, total harmonic distortion; *f*, frequency; RMS, root mean square; |*Z*|, magnitude of impedance.

A commonly used and reported impedance recording device is the nanoZ Impedance Tester developed by Plexon. The nanoZ impedance tester can record impedances for up to 64 channels, with an in-built multiplexer. By default, impedance is recorded at the single frequency of 1 kHz, outputting a single impedance value (|*Z*|) at this frequency. It should be noted that the instrument is capable of measuring over a range of frequencies and, thereby, can generate data for an impedance magnitude spectrum (half of the Bode representation). The nanoZ impedance tester measures impedance galvanostatically, using a sinusoidal perturbation at 1 nA RMS. The current amplitude is set to give a reasonable dynamic range, based on the generally expected impedance of common microelectrodes. This means electrodes that fall within this measurement window are likely to be accurately measured. Measurements at the borderline of the intended operational range will lose accuracy, and outside-of-the-range measurements may completely fail or give output values that do not reflect the interface but stem from artefacts and noise. For example, comparing impedance measurements made with the nanoZ impedance tester and with the more general electrochemistry precision instrument Autolab (Metrohm) showed that, assuming the Autolab measurement represents the true electrode impedance, the deviation with the nanoZ impedance tester was 18% at the 1 kHz point, 35% at the 10 Hz point and 300% at the 1 Hz point[59]. This deviation, particularly at lower frequencies, illustrates how the nanoZ is challenged by the electrode characteristics where the impedance increases from ~300 kΩ at the 1 kHz frequency to more than 100 MΩ at the 1 Hz point. Furthermore, the measurement in the example was performed under idealized conditions, in a Faraday cage and in a beaker of 0.01 M PBS, an ideal electrolyte. Measurements in vivo increase the challenge level.

Cables and connectors are another aspect of instrumentation that impact the measurement. The 'ideal' benchtop experiment benefits from carefully calibrated cables and measurement systems, whereas

in vivo measurements must make compromises in this respect. An accuracy contour plot (ACP) can be developed to quantify the limitations of the equipment and set-up (Fig. 4d). The ACP is typically presented on the Bode plot and defines the limits of the set-up, with the bound box representing the impedance magnitude and frequency range where the response can be accurately recorded[10]. The high impedance limit of the ACP can be determined as the impedance of the system at an open circuit whereas the lower limit represents a short circuit between the WE and CE (Fig. 4d).

Cable length has a large impact on the accuracy limit. Impedance measurements in unrestrained animals require connectors and cable arrangements with increased length and flexibility, towards lengths of 1 m. This aspect introduces errors typically manifested as parasitic capacitances, which occur between nearby connection lines with insufficient insulation between them (Fig. 4e). EIS data were recorded using an Autolab potentiostat (Metrohm) across a frequency spectrum of 1 Hz–10 kHz in three open-circuit cases: the potentiostat itself; the potentiostat and an adapter required to interface with the implant; and the potentiostat, adapter and additional cable necessary for an in vivo experiment, providing an additional 90 cm of cable length. The data highlights that as the cable length increases, the open circuit impedance decreases, due to additional parasitic capacitances in the cables. In other words, as the cable length increases, the upper boundary of the ACP decreases.

Electrophysiology systems often include built-in functionality to record impedance that can be used as an affordable, in vivo functionality test to identify electrodes that are likely to serve as good channels for electrophysiology recordings. Although several systems can measure over multiple frequencies, it is common to measure and report a single point, typically 1 kHz. Nevertheless, it is difficult to deduce, from a single-point measurement, whether the measured value has the expected features of an electrode impedance measurement and, thus, is likely to represent an actual impedance. If a spectrum is measured, it is important to recognize that deviations from true impedance are unlikely to strike all parts of the spectrum to a similar extent, as the system typically is optimized to give the best precision at the 1 kHz point. Furthermore, noise contributions are likely to impact some specific frequencies, or frequency ranges, more than others. In the user's guide of Intan Technologies RHX software, an example of an electrophysiology system with a built-in impedance tester, it is clearly specified that a "fair amount of noise and uncertainty" can be expected, a statement that needs to be considered, although is often overlooked by users.

As well as understanding the limitations of the equipment used for recording EIS data, the researcher should also critically examine the effect of the necessary additional equipment, such as cables, commutators and adapters. Although an electrode measured to have low impedance is likely to be well functioning, there can be several reasons why an electrode displays high impedance, besides failure of the electrode itself. Taken together, substantial responsibility lies on the user to reflect on these technical limitations and to not report values that are unlikely to stem from true measurements.

Interpretation of impedance spectra recorded in vivo

Alongside the physical set-up used to collect EIS data, there are important considerations to be made when interpreting and analysing impedance spectra. There are challenges entailed both in studying impedance at a single frequency, typically 1 kHz, and in drawing conclusions from full-spectrum measurements, as illustrated in the preceding section. These will here be discussed in more detail.

Limitations of studying impedance at a single frequency

It is common to simplify impedance analysis by focusing only on impedance magnitude at a single frequency, namely the 1 kHz point ($|Z|_{1\,\mathrm{kHz}}$). This is a practical approach in vivo, to reduce the total experimental time when many measurements might be required to be made on awake animals. Reporting impedance at 1 kHz has become standard practice in the field with the rationale that a nerve cell action potential occurs on the millisecond timescale, thus making impedance at 1 kHz the most important to consider for ensuring quality spike recordings. Although the relation between the low impedance of the electrode at 1 kHz and high-quality spike recordings is experimentally verified[32], it should be noted that electrophysiological signals are more complex than this, with the signal content spanning the entire frequency range[30,56,57]. Moreover, relying only on the single point 'measurement' contains an element of risk. From a single point measurement, it would be close to impossible to correctly diagnose potential electrode problems[25] and it would take additional effort to confirm linearity to differentiate between successful versus noise-contaminated measurements[21,32,60].

To highlight complexities that cannot be explained by a single point impedance measurement at 1 kHz, it is important to fully understand the relationship between $|Z|_{1\,\mathrm{kHz}}$ and recording quality. Recording quality metrics to consider include the ability to identify single spiking units, the signal-to-noise ratio of identified spikes, the spike amplitude and the array yield, that is the percentage of electrodes capable of recording discernible spikes[24,37,42,61–63]. Correlating the value of $|Z|_{1\,\mathrm{kHz}}$ with changes in the signal-to-noise ratio and the number of spiking units showed that, as the $|Z|_{1\,\mathrm{kHz}}$ value decreased over time, a decrease in the number of spiking units and in the signal-to-noise ratio of the recordings was recorded[34]. Similarly, it has been reported that over 30 weeks the array yield decreased from 95% to less than 20%, accompanied by increased impedance at 1 kHz[24]. The expected reason for this increase in impedance and subsequent decrease in recording quality is tissue encapsulation, creating an increased resistance around the interface, impeding the ability to record from nearby neurons[64]. Other mechanisms include electrode delamination or corrosion, decreasing the surface area of the electrode and therefore increasing the impedance and decreasing the recorded signal quality[38,65].

In contrast to the previous example, and highlighting the complexity of the measurement, it has also been reported that, after 3 weeks of implantation, the amplitude of the recorded spikes can decrease, along with a decrease in the impedance magnitude and a more resistive phase at 1 kHz[61]. In this case, the failure mechanism might be related to insulation degradation issues that result in an increase in the surface area of the electrodes. Exposing conductive regions of the tracks can, in some cases, result in fluid ingress into the implant and even generate current shorting between neighbouring electrodes. Through in vitro accelerated ageing of implants, it has been shown that water ingress decreases impedance across the entire frequency spectrum, accompanied by a more resistive phase at low frequencies[66]. This is characteristic of increased surface area, suggesting shorting of electrodes, which can cause a decrease in recording specificity. Therefore, the same electrodes might be recording the collective signal of a larger population of individual neurons (with respect to the original one) with poorer waveform definition[65,67]. This effect limits the ability to identify the single units and, consequently, the amplitude of the spikes[67]. Both studies highlight a challenge: failure of recording electrodes can be correlated to both increases and decreases in impedance.

To further highlight the limitations of presenting a single-point impedance measurement, we discuss a dataset that correlates EIS to different degrees of implant encapsulation[26]. Despite two impedance spectra having the same impedance magnitude at a given frequency, the phase response is different (Fig. 5). At low frequencies, the $|Z|$ value is approximately the same for the two presented groups (Fig. 5a). However, the phase and Nyquist plots (Fig. 5b,c) show that one group (dashed line) is more resistive at low frequencies. In this dataset, the solid and dashed lines represent two different degrees of histology reaction, elements of which, in week 2 as presented here, are only evident when considering the entire frequency spectrum. If in this study only $|Z|$ at a single frequency was presented, no difference would be apparent, despite clear differences across other parts of the spectrum and in the phase and Nyquist representations.

Recording impedance at 1 kHz is significantly quicker than recording impedance across a spectrum. For each recorded frequency f, the time required to generate a measurement is $1/f$. It is, however, common practice to carry out repeats at each frequency and then average the results to account for any noise or non-stationarity in the electrochemical cell. One method that has been used to decrease the time required for recording a full spectrum is the multi-sine technique[68,69]. The multi-sine technique works by applying a summation of sine waves that are multiples of the lowest frequency. The time taken for each measurement is that of the lowest applied frequency, eliminating the time required for the higher frequencies that are now measured simultaneously[68]. Although much of the work implementing the multi-sine technique is in the impedance analysis of fuel cells[68] and batteries[69], this technique has also been successfully used in in vivo interfacing electrodes[17] and is a viable option for decreasing the measurement time for full-spectrum EIS. It should, however, be noted the Kramers–Kronig relations for determining the validity of impedance data are not valid when the multi-sine technique is used[70].

The simplified recording of $|Z|$ at a single frequency can be used to provide insight into the experiment and the recording environment around the electrodes, where sudden increases and decreases may point to recording failure. However, care must be taken when drawing extended conclusions on the nature of the electrode–tissue interface based on this.

The immune response and the effect on EIS

In vivo, the environment around the electrodes is constantly altered by changes in the host tissue that arise both from implantation and the presence of the device, such as biofouling and the FBR. The insertion of a device into tissue induces mechanical trauma including bleeding, swelling and an acute inflammatory response[71–73]. The presence of an implant, in fact, has a major impact on the tissue surrounding it, and all implants trigger an immune response intended to eliminate or seal off the foreign body. Biofouling happens in the acute phase, within seconds of implantation, when proteins migrate and adsorb to the surface of the implant[74,75]. The FBR describes the body's collective immune response and tissue reactivity towards an implant[74]. Many implants cannot be eliminated from the body, so the immune system typically onsets a chronic inflammatory response[76,77]. Over time the object might become encapsulated through biofouling, protein adhesion and scar formation, effectively protecting the tissue from further direct contact[29,64,78]. Consequently, the portion of the neural tissue closest to the implant no longer functions as nerve tissue, but comprises a barrier that physically disconnects the implanted electrodes from their target neurons.

The biological mechanisms of the FBR depend on various factors such as the implant size, weight and locations[64]. Specific neural tissue types reply differently to chronic implants[73], and the degree of the FBR, such as the amount of biofouling or tissue encapsulation, depends on the specific neural interfaces and/or prosthetics[74]. In any case, the electrical properties of the tissue change over the different stages of the FBR, from acute phase to chronic inflammation, to various degrees of fibrosis, gliosis and encapsulation. The recorded impedance spectrum represents the sum of the electrode response itself and the tissue volume between the WE and CE (in a two-electrode set-up), making it challenging to disentangle if changes in the measured values relate to the electrode or its surroundings. Assuming that the WE is stable over time, EIS can be used to monitor, model and understand changes occurring at the electrode–tissue interface and, to some extent, also discern reactions and remodelling occurring in the surrounding tissue[19,22,25,26,79].

Impedance spectra and histological changes, representing the degree of FBR and tissue encapsulation of the device, have been correlated[17,22,26]. For example, it has been possible to determine the extent of the tissue response through histology, with differences

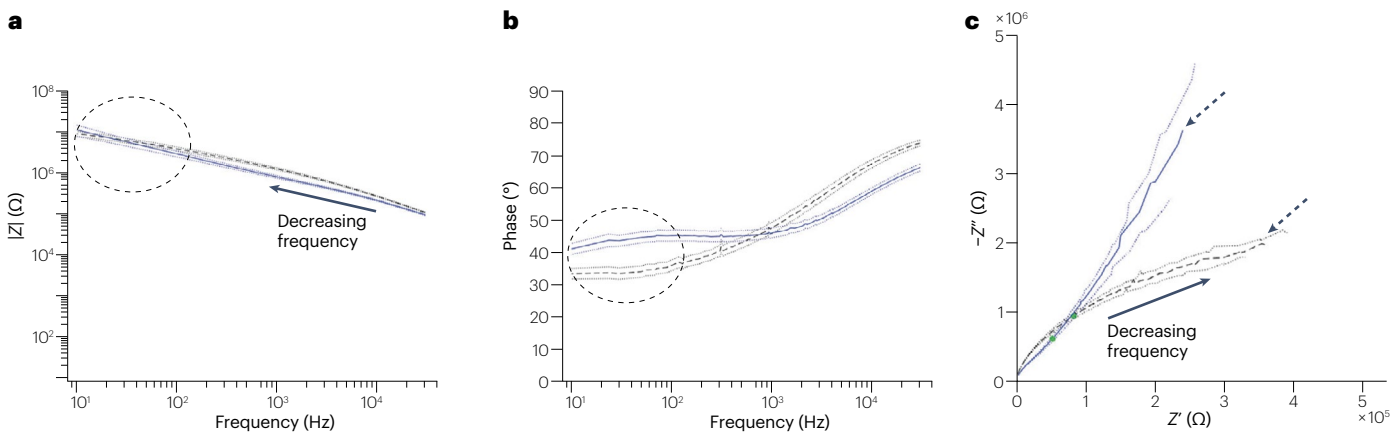

Fig. 5 | Interpretation of impedance data. a, Impedance magnitude ($|Z|$) across the frequency spectrum for two characteristic encapsulation types. The dotted circle highlights the low-frequency region where the $|Z|$ value for both electrode types is similar. **b**, The phase plot across the frequency spectrum for to the same dataset presented in **a**. The dotted circle highlights the low-frequency region. **c**, The Nyquist representation of the same dataset, with arrows pointing to the data recorded at the lowest frequency. Z', real component of impedance; Z'', imaginary component of impedance. Parts **a**–**c** are adapted with permission from ref. 26, Elsevier.

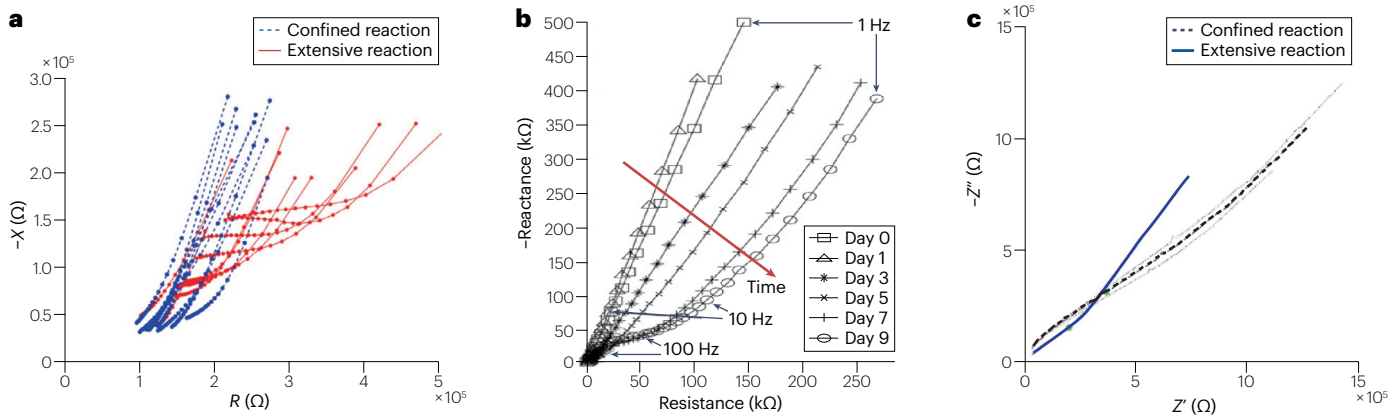

Fig. 6 | Impedance spectra as Nyquist plots representing changes in tissue impedance. a, Impedance spectra of all electrodes 7 days after implantation, with red data representing an extensive histological reaction and blue dashed lines a smaller/contained reaction. Part **a** is adapted with permission from ref. 22, IOP. **b**, Impedance spectra of one representative electrode for 9 days post implantation. Part **b** is adapted with permission from ref. 17, IOP. **c**, Average impedance spectra of all electrodes after 12 weeks of implantation, with the blue line representing an extensive histological reaction and the black dashed line a confined reaction. Part **c** is adapted with permission from ref. 26, Elsevier. Z', real component of impedance; Z'', imaginary component of impedance.

apparent after 5 days[22]. Extensive tissue reactions were found to be correlated to an increase in the resistive component of impedance, indicated by a shift to the right in the Nyquist plot (Fig. 6a). In a different study, an increase in the resistive component of impedance was identified at 12 weeks, along with the appearance of a semicircular arc in the Nyquist plot from 5 days[17], similarly taken as an indication of emerging tissue response (Fig. 6b). By contrast, it has also been reported that the resistive component was reduced for smaller, confined tissue reactions, compared with larger and extensive tissue changes[26]. Furthermore, the presence of the semicircular arc in the Nyquist representation was greater for smaller, confined reactions (Fig. 6c). This contradiction can be rationalized by two observations. First, the encapsulation tissue has a high collagen content, which is less resistive than the highly cellular and myelin-rich brain tissue. Second, the fibrotic tissue growth causes the array base to be pushed up, resulting in the formation of fluid-filled gaps that create less resistive pathways for current spread so that the more resistive encapsulating layer is not included.

The contradicting information from the studies above highlights the complexity of establishing a generalizable method to reconstruct the tissue–electrode electrochemical elements from a single dataset of impedance. A useful interpretation of EIS data must consider various factors that are specific to each experimental set-up. First, one must consider the placement of the WE and CE relative to both the neural tissue of interest and the tissue encapsulation. It might be the case that tissue reactions result in new, low-resistive current pathways from the WE to CE, where the encapsulating tissue between electrodes and signalling neurons is not included[26,80]. Second, the overall performance of neural implants varies depending on the implant materials and the surrounding tissue, with differences even present between brain regions, where one might assume some level of similarity[81]. Third, within the tissue itself, impedance depends on many factors, such as the degree of cell adhesion, the kinetics of cell growth, neuronal differentiation, cell apoptosis and neural degeneration[21]. Fourth, the type and layout of the electrode array is decisive for how cells interact with the device and the electrodes, specifically if cells are able to adhere directly to the electrode surface. Electrode arrays may feature protruding electrodes, electrodes patterned on the outermost surface of the array or electrodes recessed in an electrically insulating layer, three variations which change the nature of the electrode–tissue contact especially for microelectrodes[82,83]. If the size of the electrode is small relative to the typical cell body, less than 100 μm, it is unlikely that cells will adhere directly to the recessed electrode surface if the aspect ratio between the recess depth and the electrode diameter is high[84,85]. Despite these differences, within a specific experiment, the trend correlating impedance to histology is reproducible across animals, highlighting the value of impedance as a metric to decode and quantify tissue changes around a neural implant[17,22,26]. Thus, impedance to quantify tissue changes can be reproducible under the condition that implants are compared within the same category, tissue, WE and CE location.

To separate and understand the tissue changes further, electrical circuit modelling might be used. However, there are limitations in using circuit models to aid understanding (see Supplementary information 1). For example, different models might give the same results[4]. At the same time, the same model used to fit different datasets might give opposing results because of the different histology of the targets and different experiment configurations (such as the CE location)[22,26]. In 2024, it was demonstrated that the effects of tissue encapsulation can be partially mitigated by specific stimulation protocols, effectively restoring electrode properties to near baseline levels[86]. This work rationalizes the rejuvenation through a change in the 'effective thickness' of the encapsulation layer, where the 'effective encapsulation thickness' is calculated through an equivalent circuit model and represents an estimation of the tissue encapsulation around the electrodes. The underlying changes of the encapsulation layers should be studied, for example using histology or scanning electron microscopy, to relate changes in 'effective thickness' to the physical size and properties of the encapsulation layer. Electrical stimulation has been proposed to temporarily reduce in vivo impedance and rejuvenate an interface[17,32,37,43,87–90] (Supplementary Information, section 2).

Outlook

EIS measurements can, within a specific implant and study, give useful practical outputs to inform stimulation or recording paradigms and future implant designs. However, due to diverse tissue

properties and tissue responses, coupled with varying technical implementations of the implants, quantifiable results cannot readily be exported across studies. Nonetheless, when methods such as the electrochemical set-up and assumptions are comprehensively reported, the data holds value in delineating trends that persist across different groups and studies. Some suggestions can be made to facilitate more transparent and informative reporting of impedance data in the future.

Clear communication of the electrochemical set-up, including details such as the mode (potentiostatic or galvanostatic), amplitude, frequency range and electrochemical cell configuration (two-electrode or three-electrode set-up), is crucial for meaningful interpretation of impedance measurements[91]. Furthermore, it is essential to note the placement of electrodes with respect to each other and with respect to any encapsulating tissue. If electrical circuit modelling is used to support interpretation, it should be kept in mind that by adding more circuit elements to a model, it is highly likely that a model can be forced to fit the data. Conversely, a model fit does not guarantee that the model components introduced have any physical correlation or meaning. This is a limitation of modelling. To use extracted model subcomponents to draw conclusions about, for example, the state of the FBR, one first needs to ensure validity of the model itself to the specific situation.

In general, the presentation of impedance across the frequency spectrum can provide many insights into failure mechanisms and changes at the electrode and within the tissue, whereas single-frequency measurements should be used cautiously. We recognize that this may be challenging both in terms of time constraints to generate these data for large numbers of electrodes in awake animals and in the presentation of data, when presenting measurements for many electrodes over days or weeks. In instances where a single value, such as $|Z|_{1\,\mathrm{kHz}}$, is presented, the validity of the measurement should ideally be supported by collecting the entire frequency spectrum at least on certain occasions to ensure this represents realistic electrode characteristics, or through a Lissajous plot.

Consistency in terminology is another aspect where a clear consensus in the field has been lacking. Although impedance spectroscopy is a purely electrochemical technique with clearly defined electrode terminology, the implementation in the animal model typically means that the electrodes serve dual roles leading to ambiguous electrode naming. The electrochemical naming with WE and CE clearly notes which electrode is primarily investigated (working) versus the electrode that merely serves for closing the circuit (counter). The electrochemical RE is passive in this circuit. When this terminology is used in electrophysiology this leads to confusion, as the RE in this sense typically refers to the electrode connected as a reference signal in the differential amplifier, sometimes in addition noted as 'ground'. At the same time, the ground electrode is commonly given the role of closing the circuit, in principle then serving as the CE. This ambiguity highlights the need to standardize electrode naming. We propose that reporting of impedance spectroscopy measurements should clarify which electrodes served as the WE and CE, and if a third RE was used to form a three-electrode configuration, to specifically refer to this as the 'electrochemical RE' to differentiate from the 'recording reference'.

Within the experimental set-up, perhaps the largest opportunity to shape the field lies in complementing the classical two-electrode measurement with additional impedance measurement configurations. One of the major challenges in interpreting in vivo impedance data is that it is generally not possible, with certainty, to differentiate between changes occurring within the electrode array, immediately at the electrode surface, or changes related to tissue between the WE and CE. The development of a stable on-the-probe electrochemical cell configuration, including an electrochemical RE, would enable robust continuous three-electrode measurements in vivo over time. The adoption of this configuration would simplify the separation of the contribution of the electrode/electrode surface to impedance changes from those related to tissue. Some steps towards an on-the-probe RE have been taken[33], but more research and experiments are necessary to establish a translated three-electrode electrochemical cell. Furthermore, the probe shank could include additional monitoring elements, such as impedance circuits reporting on the integrity of the probe material layers, to further differentiate between changes within the device (for example, short circuits within the probe) and reactions at the electrode surface. Combining a broader set of impedance measurement options, more information could be extracted from the in vivo impedance experiment itself.

By changing the way impedance data are presented and discussed, there is the potential for this measurement to be a meaningful and respected metric, as opposed to a dataset generated due to expectation, without clear interpretation. With these recommendations, it is a promising thought that a larger body of data, measured in a consistent and well-reported manner, could support the development of new mathematical modelling tools. Importantly, impedance stands as the primary method enabling the assessment of electrode properties in vivo using the same electrode arrays as for the electrophysiological experiments, without additional interventions, underscoring its significance in the field of neural interface research.

Published online: 3 January 2025

References

1. Hatsopoulos, N. G. & Donoghue, J. P. The science of neural interface systems. *Annu. Rev. Neurosci.* **32**, 249–266 (2009).
2. Macdonald, D. D. Reflections on the history of electrochemical impedance spectroscopy. *Electrochim. Acta* **51**, 1376–1388 (2006).
3. Orazem, M. E. & Tribollet, B. A tutorial on electrochemical impedance spectroscopy. *ChemTexts* **6**, 12 (2020).
4. Macdonald, J. R. Impedance spectroscopy. *Ann. Biomed. Eng.* **20**, 289–305 (1992).
5. Kemp, N. T. A tutorial on electrochemical impedance spectroscopy and nanogap electrodes for biosensing applications. *IEEE Sens. J.* **21**, 22232–22245 (2021).
6. Gabrielli, C. Once upon a time there was EIS. *Electrochim. Acta* **331**, 135324 (2020).
7. Amini, M., Hisdal, J. & Kalvoy, H. Applications of bioimpedance measurement techniques in tissue engineering. *J. Electr. Bioimpedance* **9**, 142–158 (2018).
8. Macdonald, J. R. & Johnson, W. B. in *Impedance Spectroscopy* 1–20 (Wiley, 2018).
9. Wang, S. et al. Electrochemical impedance spectroscopy. *Nat. Rev. Methods Primers* **1**, 41 (2021).
   This review discusses the impedance spectroscopy experiment, providing a strong foundation for any application.
10. Lazanas, A. C. & Prodromidis, M. I. Electrochemical impedance spectroscopy—a tutorial. *ACS Meas. Sci. Au* **3**, 162–193 (2023).
11. Boukamp, B. A. A linear Kronig–Kramers transform test for immittance data validation. *J. Electrochem. Soc.* **142**, 1885 (1995).
12. Orazem, M. E., Pébère, N. & Tribollet, B. Enhanced graphical representation of electrochemical impedance data. *J. Electrochem. Soc.* **153**, B129 (2006).
13. Magar, H. S., Hassan, R. Y. A. & Mulchandani, A. Electrochemical impedance spectroscopy (EIS): principles, construction, and biosensing applications. *Sensors* **21**, 6578 (2021).
14. Cogan, S. F. Neural stimulation and recording electrodes. *Annu. Rev. Biomed. Eng.* **10**, 275–309 (2008).
15. Cogan, S. F., Troyk, P. R., Ehrlich, J., Gasbarro, C. M. & Plante, T. D. The influence of electrolyte composition on the in vitro charge-injection limits of activated iridium oxide (AIROF) stimulation electrodes. *J. Neural Eng.* **4**, 79 (2007).
16. Bobacka, J., Lewenstam, A. & Ivaska, A. Electrochemical impedance spectroscopy of oxidized poly(3,4-ethylenedioxythiophene) film electrodes in aqueous solutions. *J. Electroanalytical Chem.* **489**, 17–27 (2000).
17. Lempka, S. F., Miocinovic, S., Johnson, M. D., Vitek, J. L. & McIntyre, C. C. In vivo impedance spectroscopy of deep brain stimulation electrodes. *J. Neural Eng.* **6**, 046001 (2009).
18. Weiland, J. D. & Anderson, D. J. Chronic neural stimulation with thin-film, iridium oxide electrodes. *IEEE Trans. Biomed. Eng.* **47**, 911–918 (2000).

19. Mercanzini, A., Colin, P., Bensadoun, J. C., Bertsch, A. & Renaud, P. In vivo electrical impedance spectroscopy of tissue reaction to microelectrode arrays. *IEEE Trans. Biomed. Eng.* **56**, 1909–1918 (2009).
20. Shinwari, M. W. et al. Microfabricated reference electrodes and their biosensing applications. *Sensors* **10**, 1679–1715 (2010).
21. Krukiewicz, K. Electrochemical impedance spectroscopy as a versatile tool for the characterization of neural tissue: a mini review. *Electrochem. Commun.* **116**, 106742 (2020).
22. Williams, J. C., Hippensteel, J. A., Dilgen, J., Shain, W. & Kipke, D. R. Complex impedance spectroscopy for monitoring tissue responses to inserted neural implants. *J. Neural Eng.* **4**, 410 (2007).
This study analyses full spectrum impedance recorded in vivo and provides evidence of tissue impedance represented by a semicircular arc at high frequencies.
23. Otte, E., Vlachos, A. & Asplund, M. Engineering strategies towards overcoming bleeding and glial scar formation around neural probes. *Cell Tissue Res.* **387**, 461–477 (2022).
This review discusses different neural probes and the biological challenges of implantation and maintenance of the devices on chronic timescales.
24. Joshi-Imre, A. et al. Chronic recording and electrochemical performance of amorphous silicon carbide-coated Utah electrode arrays implanted in rat motor cortex. *J. Neural Eng.* **16**, 046006 (2019).
25. Straka, M. M., Shafer, B., Vasudevan, S., Welle, C. & Rieth, L. Characterizing longitudinal changes in the impedance spectra of in-vivo peripheral nerve electrodes. *Micromachines* **9**, 587 (2018).
26. Cody, P. A., Eles, J. R., Lagenaur, C. F., Kozai, T. D. Y. & Cui, X. T. Unique electrophysiological and impedance signatures between encapsulation types: an analysis of biological Utah array failure and benefit of a biomimetic coating in a rat model. *Biomaterials* **161**, 117–128 (2018).
This manuscript evaluates full spectrum impedance measurements alongside histology, discussing the effect of different lead wire encapsulation types.
27. Ward, M. P., Rajdev, P., Ellison, C. & Irazoqui, P. P. Toward a comparison of microelectrodes for acute and chronic recordings. *Brain Res.* **1282**, 183–200 (2009).
28. Malaga, K. A. et al. Data-driven model comparing the effects of glial scarring and interface interactions on chronic neural recordings in non-human primates. *J. Neural Eng.* **13**, 016010 (2016).
29. Barrese, J. C. et al. Failure mode analysis of silicon-based intracortical microelectrode arrays in non-human primates. *J. Neural Eng.* **10**, 066014 (2013).
30. Ludwig, K. A., Uram, J. D., Yang, J., Martin, D. C. & Kipke, D. R. Chronic neural recordings using silicon microelectrode arrays electrochemically deposited with a poly(3,4-ethylenedioxythiophene) (PEDOT) film. *J. Neural Eng.* **3**, 59–70 (2006).
31. Vetter, R. J., Williams, J. C., Hetke, J. F., Nunamaker, E. A. & Kipke, D. R. Chronic neural recording using silicon-substrate microelectrode arrays implanted in cerebral cortex. *IEEE Trans. Biomed. Eng.* **51**, 896–904 (2004).
32. Bodart, C. et al. Electropolymerized poly(3,4-ethylenedioxythiophene) (PEDOT) coatings for implantable deep-brain-stimulating microelectrodes. *ACS Appl. Mater. Interfaces* **11**, 17226–17233 (2019).
33. Boehler, C. et al. Actively controlled release of dexamethasone from neural microelectrodes in a chronic in vivo study. *Biomaterials* **129**, 176–187 (2017).
34. Vomero, M. et al. On the longevity of flexible neural interfaces: establishing biostability of polyimide-based intracortical implants. *Biomaterials* **281**, 121372 (2022).
35. Hughes, C. L. et al. Neural stimulation and recording performance in human sensorimotor cortex over 1500 days. *J. Neural Eng.* **18**, 045012 (2021).
36. Wang, L. et al. Self-spreadable octopus-like electrode arrays for long-term neural recordings. *Acta Phys. Sin.* **36**, 1909035 (2020).
37. Cassar, I. R. et al. Electrodeposited platinum–iridium coating improves in vivo recording performance of chronically implanted microelectrode arrays. *Biomaterials* **205**, 120–132 (2019).
38. Sankar, V. et al. Electrode impedance analysis of chronic tungsten microwire neural implants: understanding abiotic vs. biotic contributions. *Front. Neuroeng.* **7**, 13 (2014).
39. Prasad, A. et al. Abiotic–biotic characterization of Pt/Ir microelectrode arrays in chronic implants. *Front. Neuroeng.* **7**, 2 (2014).
40. Wang, C. et al. Characteristics of electrode impedance and stimulation efficacy of a chronic cortical implant using novel annulus electrodes in rat motor cortex. *J. Neural Eng.* **10**, 046010 (2013).
41. McConnell, G. C., Butera, R. J. & Bellamkonda, R. V. Bioimpedance modeling to monitor astrocytic response to chronically implanted electrodes. *J. Neural Eng.* **6**, 055005 (2009).
42. Prasad, A. & Sanchez, J. C. Quantifying long-term microelectrode array functionality using chronic in vivo impedance testing. *J. Neural Eng.* **9**, 026028 (2012).
43. Torres-Martinez, N. et al. Reliability of parylene-based multi-electrode arrays chronically implanted in adult rat brains, and evidence of electrical stimulation on contact impedance. *J. Neural Eng.* **16**, 066047 (2019).
44. Purcell, E. K., Thompson, D. E., Ludwig, K. A. & Kipke, D. R. Flavopiridol reduces the impedance of neural prostheses in vivo without affecting recording quality. *J. Neurosci. Methods* **183**, 149–157 (2009).
45. Kane, S. R. et al. Electrical performance of penetrating microelectrodes chronically implanted in cat cortex. *IEEE Trans. Biomed. Eng.* **60**, 2153–2160 (2013).
46. Cvancara, P. et al. Stability of flexible thin-film metallization stimulation electrodes: analysis of explants after first-in-human study and improvement of in vivo performance. *J. Neural Eng.* **17**, 046006 (2020).
47. Seaton, B. T. & Heien, M. L. Biocompatible reference electrodes to enhance chronic electrochemical signal fidelity in vivo. *Anal. Bioanal. Chem.* **413**, 6689–6701 (2021).
This review highlights the challenges of chronic in vivo electrochemical experiments and provides options and perspectives on stable REs for improved signals.
48. Moussy, F. & Harrison, D. J. Prevention of the rapid degradation of subcutaneously implanted Ag/AgCl reference electrodes using polymer coatings. *Anal. Chem.* **66**, 674–679 (1994).
49. Hashemi, P. et al. Chronically implanted, nafion-coated Ag/AgCl reference electrodes for neurochemical applications. *ACS Chem. Neurosci.* **2**, 658–666 (2011).
50. Jackson, W. F. & Duling, B. R. Toxic effects of silver–silver chloride electrodes on vascular smooth muscle. *Circulation Res.* **53**, 105–108 (1983).
51. Robbins, E. M., Castagnola, E. & Cui, X. T. Accurate and stable chronic in vivo voltammetry enabled by a replaceable subcutaneous reference electrode. *iScience* **25**, 104845 (2022).
52. Zhang, F. et al. Reference and counter electrode positions affect electrochemical characterization of bioanodes in different bioelectrochemical systems. *Biotechnol. Bioeng.* **111**, 1931–1939 (2014).
53. Yang, H. et al. An iridium oxide reference electrode for use in microfabricated biosensors and biochips. *Lab Chip* **4**, 42–46 (2004).
54. Fan, B. et al. Flexible, diamond-based microelectrodes fabricated using the diamond growth side for neural sensing. *Microsyst. Nanoeng.* **6**, 42 (2020).
55. Bard, A. J., Faulkner, L. R. & White, H. S. *Electrochemical Methods: Fundamentals and Applications* 3rd edn (Wiley, 2022).
56. Orazem, M. E. & Tribollet, B. in *Electrochemical Impedance Spectroscopy* (eds Orazem, M. E. & Tribollet, B.) 165–189 (Wiley, 2017).
57. Orazem, M. E. & Tribollet, B. in *Electrochemical Impedance Spectroscopy* (eds Orazem, M. E. & Tribollet, B.) 139–164 (Wiley, 2017).
58. Mierzejewski, M., Steins, H., Kshirsagar, P. & Jones, P. D. The noise and impedance of microelectrodes. *J. Neural Eng.* **17**, 052001 (2020).
59. Boehler, C., Carli, S., Fadiga, L., Stieglitz, T. & Asplund, M. Tutorial: guidelines for standardized performance tests for electrodes intended for neural interfaces and bioelectronics. *Nat. Protoc.* **15**, 3557–3578 (2020).
This tutorial-style review provides guidelines for a range of standardized electrochemical tests, including EIS, for electrodes to be used at neural interfaces.
60. Goh, J., Eluagu, C., Babauta, J. & Orazem, M. Comparison of approaches for assessing linearity of impedance measurements. *J. Electrochem. Soc.* **171**, 036508 (2024).
61. Liu, X. et al. Stable, long-term single-neuronal recording from the rat spinal cord with flexible carbon nanotube fiber electrodes. *J. Neural Eng.* **19**, 056024 (2022).
62. Jiang, J., Willett, F. R. & Taylor, D. M. Relationship between microelectrode array impedance and chronic recording quality of single units and local field potentials. In *36th Annual International Conf. IEEE Engineering in Medicine and Biology Society* 3045–3048 (IEEE, 2014).
63. Nolta, N. F., Christensen, M. B., Crane, P. D., Skousen, J. L. & Tresco, P. A. BBB leakage, astrogliosis, and tissue loss correlate with silicon microelectrode array recording performance. *Biomaterials* **53**, 753–762 (2015).
64. Wellman, S. M. et al. A materials roadmap to functional neural interface design. *Adv. Funct. Mater.* **28**, 1701269 (2018).
65. Jorfi, M., Skousen, J. L., Weder, C. & Capadona, J. R. Progress towards biocompatible intracortical microelectrodes for neural interfacing applications. *J. Neural Eng.* **12**, 011001 (2015).
66. Takmakov, P. et al. Rapid evaluation of the durability of cortical neural implants using accelerated aging with reactive oxygen species. *J. Neural Eng.* **12**, 026003 (2015).
67. Lewis, C. M. et al. Recording quality is systematically related to electrode impedance. *Adv. Healthc. Mat.* **13**, 2303401 (2024).
68. Ulgut, B. Methods-employing multisine electrochemical impedance spectroscopy for batteries in galvanostatic mode. *J. Electrochem. Soc.* **169**, 110510 (2022).
69. Fortin, P., Gerhardt, M. R., Ulleberg, Ø., Zenith, F. & Holm, T. Multi-sine EIS for early detection of PEMFC failure modes. *Front. Energy Res.* **10**, 855985 (2022).
70. You, C., Zabara, M. A., Orazem, M. E. & Ulgut, B. Application of the Kramers–Kronig relations to multi-sine electrochemical impedance measurements. *J. Electrochem. Soc.* **167**, 020515 (2020).
71. Kozai, T. D., Vazquez, A. L., Weaver, C. L., Kim, S. G. & Cui, X. T. In vivo two-photon microscopy reveals immediate microglial reaction to implantation of microelectrode through extension of processes. *J. Neural Eng.* **9**, 066001 (2012).
72. Tresco, P. A. & Winslow, B. D. The challenge of integrating devices into the central nervous system. *Crit. Rev. Biomed. Eng.* **39**, 29–44 (2011).
73. Polikov, V. S., Tresco, P. A. & Reichert, W. M. Response of brain tissue to chronically implanted neural electrodes. *J. Neurosci. Methods* **148**, 1–18 (2005).
74. Carnicer-Lombarte, A., Chen, S.-T., Malliaras, G. G. & Barone, D. G. Foreign body reaction to implanted biomaterials and its impact in nerve neuroprosthetics. *Front. Bioeng. Biotechnol.* **9**, 622524 (2021).
75. Bixler, G. & Bhushan, B. Review article: Biofouling: lessons from nature. *Philos. Trans. A Math. Phys. Eng. Sci.* **370**, 2381–2417 (2012).
76. Biran, R., Martin, D. C. & Tresco, P. A. Neuronal cell loss accompanies the brain tissue response to chronically implanted silicon microelectrode arrays. *Exp. Neurol.* **195**, 115–126 (2005).
77. McConnell, G. C. et al. Implanted neural electrodes cause chronic, local inflammation that is correlated with local neurodegeneration. *J. Neural Eng.* **6**, 056003 (2009).

78. Usoro, J. O., Sturgill, B. S., Musselman, K. C., Capadona, J. R. & Pancrazio, J. J. Intracortical microelectrode array unit yield under chronic conditions: a comparative evaluation. *Micromachines* **12**, 972 (2021).
79. Meijs, S. et al. Influence of fibrous encapsulation on electro-chemical properties of TiN electrodes. *Med. Eng. Phys.* **38**, 468–476 (2016).
80. Woeppel, K. et al. Explant analysis of Utah electrode arrays implanted in human cortex for brain–computer interfaces. *Fronti. Bioeng. Biotechnolol.* **9**, 759711 (2021).
81. Evers, J. & Lowery, M. The active electrode in the living brain: the response of the brain parenchyma to chronically implanted deep brain stimulation electrodes. *Operative Neurosurg.* **20**, 131–140 (2021).
82. Kilias, A. et al. Intracortical probe arrays with silicon backbone and microelectrodes on thin polyimide wings enable long-term stable recordings in vivo. *J. Neural Eng.* **18**, 066026 (2021).
83. Steins, H. et al. A flexible protruding microelectrode array for neural interfacing in bioelectronic medicine. *Microsyst. Nanoeng.* **8**, 131 (2022).
84. Zhou, T. et al. Syringe-injectable mesh electronics integrate seamlessly with minimal chronic immune response in the brain. *Proc. Natl Acad. Sci. USA* **114**, 5894–5899 (2017).
85. Ferguson, M., Sharma, D., Ross, D. & Zhao, F. A critical review of microelectrode arrays and strategies for improving neural interfaces. *Adv. Healthc. Mater.* **8**, e1900558 (2019).
86. O'Sullivan, K. P., Orazem, M. E., Otto, K. J., Butson, C. R. & Baker, J. L. Electrical rejuvenation of chronically implanted macroelectrodes in nonhuman primates. *J. Neural Eng.* **21**, 036056 (2024).
87. Otto, K. J., Johnson, M. D. & Kipke, D. R. Voltage pulses change neural interface properties and improve unit recordings with chronically implanted microelectrodes. *IEEE Trans. Biomed. Eng.* **53**, 333–340 (2006).
88. Johnson, M. D., Otto, K. J. & Kipke, D. R. Repeated voltage biasing improves unit recordings by reducing resistive tissue impedances. *IEEE Trans. Neural Syst. Rehabil. Eng.* **13**, 160–165 (2005).
89. Evers, J. et al. Stimulation-induced changes at the electrode–tissue interface and their influence on deep brain stimulation. *J. Neural Eng.* **19**, 046004 (2022).
90. Hemm, S. et al. Evolution of brain impedance in dystonic patients treated by GPi electrical stimulation. *Neuromodulation* **7**, 67–75 (2004).
91. Minteer, S. et al. New guidelines for presenting electrochemical data in all ACS journals. *J. Org. Chem.* **88**, 4036–4037 (2023).
92. Vomero, M. et al. Conformable polyimide-based μECoGs: bringing the electrodes closer to the signal source. *Biomaterials* **255**, 120178 (2020).

Acknowledgements

This work was supported by the CatWalk Spinal Cord Injury Trust and the Health Research Council of New Zealand (project grant and HRC/Catwalk Partnership 19/895). This work was also supported by the Assistant Secretary of Defense for Health Affairs endorsed by the Department of Defense (US $534,258) through the Spinal Cord Injury Research Program (award HT9425-23-1-0492). Opinions, interpretations, conclusions and recommendations are those of the authors and are not necessarily endorsed by the Department of Defense. B. Hazelgrove was supported by a University of Auckland Doctoral Scholarship. B.R. was supported by the Neurological Foundation First Fellowship (1952 FF). M.A. and L.M were supported by the European Research Council (ERC) under the European Union's Horizon 2020 Research and Innovation programme (grant agreement 101113487 'ProBER'), the European Innovation Council H2020 programme (grant agreements 899287 'NeuraViper' and 101099366 'BioFINE'), and the Chalmers Gender Initiative for Excellence (GENIE).

Author contributions

B. Hazelgrove and L.M. researched data for the article. All authors substantially contributed to the discussion of the content, wrote the manuscript, and reviewed and edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s44287-024-00126-6.

Peer review information *Nature Reviews Electrical Engineering* thanks Alexandra Joshi-Imre, who co-reviewed with Yupeng Wu, Katarzyna Krukiewicz and Mark E. Orazem for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Related links

Blackrock Neurotech: https://blackrockneurotech.com/
Intan Technologies: https://intantech.com/
Metrohm: https://www.metrohm.com/en_au/products/electrochemistry.html
nanoZ Impedance Tester: https://plexon.com/products/nanoz-impedance-tester/

© Springer Nature Limited 2025